

Aditya Khan

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EDUCATION

Columbia University
Ph.D. Student – Statistics

Sept. 2026 – Present
New York, NY

University of Toronto – St. George Campus (Trinity College)
Hons. BSc – Data Science (Specialist), Statistics (Specialist), Computer Science (Minor)

- Cumulative GPA: 3.95/4.00; Enrolled in Arts and Science Internship Program (Co-op).
- Supervisors: Meredith Franklin (Statistics), Anna Goldenberg (Computer Science)

Sept. 2021 – Jun. 2026
Toronto, ON

PUBLICATIONS

Dynamics of an SEIRV Epidemic Model: Effects of Reinfection

Aditya Khan, Lin Wang, Juxin Liu, Momiao Xiong.
Journal of Biological Dynamics.

Enhancing Mental Health Counseling Support in Bangladesh using Culturally-Grounded Knowledge

Md Arif Hasan, Azhagu Meena SP, Aditya Khan, Abu Md Akteruzzaman Bhuiyan, Helal Uddin Ahmed, Joysree Debi, Farig Sadeque, En-Shiun Annie Lee and Syed Ishtiaque Ahmed.
CLPsych Workshop 2026 at ACL 2026.

Dynamic Meta-Metrics: Source-Sentence Conditioned Weighting for MT Evaluation

Luke Zhang, Justin Vasselli, Aditya Khan, York Hay Ng, En-Shiun Annie Lee.
Association for Computational Linguistics (ACL) SRW 2026.

Simple Additions, Substantial Gains: Expanding Scripts, Languages, and Lineage Coverage in URIEL+

Mason Shipton, York Hay Ng, Aditya Khan, Phuong Hanh Hoang, Xiang Lu, A. Seza Doğruöz, En-Shiun Annie Lee.
Language Resources and Evaluation Conference (LREC) 2026.

Modality Matching Matters: Calibrating Language Distances for Cross-Lingual Transfer in URIEL⁺

York Hay Ng^{}, Aditya Khan^{*}, Xiang Lu^{*}, Matteo Salloum, Michael Zhou, Phuong Hanh Hoang, A. Seza Doğruöz, En-Shiun Annie Lee.*
European Chapter of the Association for Computational Linguistics (EACL) SRW 2026 (Spotlight).

URIEL+: Enhancing Linguistic Inclusion and Usability in a Typological and Multilingual Knowledge Base⁺

Aditya Khan^{}, Mason Shipton^{*}, David Anugraha, Kaiyao Duan, Phuong Hanh Hoang, Eric Khiu, A. Seza Doğruöz, En-Shiun Annie Lee.*
International Conference on Computational Linguistics (COLING) 2025.

A Faculty Initiative Addressing Gender Disparity at a Small STEM-Focused University: A Case Study⁺

Amane Takeuchi^{}, Aditya Khan^{*}, Phuong Hanh Hoang^{*}, Jian Yun Zhuang, Randy J. Fortier, Mariana Shimabukuro, Michael Miljanovic, En-Shiun Annie Lee.*
ACM Virtual Global Computing Education Conference (SIGSCE Virtual) 2024.

Toward A Reinforcement-Learning-Based System for Adjusting Medication to Minimize Speech Disfluency

Pavlos Conostas^{}, Vikram Rawal^{*}, Matthew Honorio Oliveira^{*}, Andreas Conostas^{*}, Aditya Khan^{*}, Kaison Cheung^{*}, Najma Sultani^{*}, Carrie Chen^{*}, Micol Altomare^{*}, Michael Akzam^{*}, Jiacheng Chen^{*}, Vhea He^{*}, Lauren Altomare^{*}, Heraa Muqri^{*}, Asad Khan^{*}, Nimit Amikumar Bhansali^{*}, Youssef Rachad^{*}, Michael Guerzhoy.*
Machine Learning for Cognitive and Mental Health Workshop (ML4CMH) at AAAI 2024.

^{*} Indicates equal contribution.

⁺ Indicates that I also presented this paper myself at the associated conference.

PRESENTATIONS WITHOUT ARCHIVAL PROCEEDINGS

- [T] **SSC Annual Meeting** Jun. 2026
- Adaptive Confidence Sets for Sobolev Regular Densities Under Local Differential Privacy
- [T] **Canadian Statistics Student Conference (CSSC) 2026** May 2026
- RESAPLE: An Approximate One-Step Restricted Likelihood Estimator of Spatial Dependence for Exploratory Spatial Analysis
- [T] **SIGTYP Workshop at EACL 2026** Mar. 2026
- Modality Matching Matters: Calibrating Language Distances for Cross-Lingual Transfer in Multilingual NLP
- [P] **Applied Research in Action (ARIA) Showcase 2025** Nov. 2025
- Modality Matching Matters: Calibrating Language Distances for Cross-Lingual Transfer in Multilingual NLP
- [T] **The Third Joint SIAM/CAIMS Annual Meetings (AN25)** Jul. 2025
- Endemicity and Eradication of Diseases in the Presence of Vaccination and Reinfection in a SEIRV Model
- [T] **SSC Annual Meeting 2025** May 2025
- Comparing Exact and Approximate Inference for Spatial Autocorrelation Tests: A Power Analysis
- [T] **Canadian Statistics Student Conference (CSSC) 2025** May 2025
- Crowds, Chaos, and Commutes: Making Inference on the Spatiotemporal Impact of Special Events on Travel Times
- [T] **Trinity College Undergraduate Research Conference (TCURC) 2025** Mar. 2025
- Endemicity and Eradication of Diseases in the Presence of Vaccination and Reinfection in a SEIRV Model
- Key: [P] = Poster, [T] = Oral.

PREPRINTS AND PAPERS UNDER REVIEW

[P] **RESAPLE: An Approximate One-Step Restricted Likelihood Estimator of Spatial Dependence for Exploratory Spatial Analysis**

Aditya Khan, Meredith Franklin. Under Review at Spatial Statistics.

[M] **A Human-Centred Evaluation of Language Distance Measures in Multilingual NLP**

Aditya Khan, York Hay Ng, Madelaine P. Jarcew, En-Shiun Annie Lee. Under Review at ACL ARR.

[M] **Explainable Typological Feature Prediction with Large Language Models: An In-Context Learning Approach**

Qianwen Wang, York Hay Ng*, Aditya Khan, En-Shiun Annie Lee. Under Review at ACL ARR.*

- We show that it is possible to attain explainable, SOTA performance on typological feature prediction with LLMs.

[M] **Greater Than the Sum of Its Parts: When Language Distances are Practical for Equitable Cross-Lingual Transfer**

Razan Ahsan Rifandi, York Hay Ng*, Aditya Khan*, En-Shiun Annie Lee. Under Review at ACL ARR.*

- We show that language-distance based transfer language rankers are still performant and resource-level equitable.

[M] **Parametric Hotspot Analysis with the Getis-Ord Statistic**

Aditya Khan, Ian Zhang. Under Review at Spatial Statistics.

- We develop a parametric hotspot detection framework for the Getis-Ord statistic, including under residualisation.

Key: [P] = Preprint on ArXiv (hyperlinked), [M] = Manuscript submitted and available on request.

* Indicates equal contribution.

WORK EXPERIENCE

ML Research Intern

May 2026 – Aug. 2026

Vector Institute

- Developing unsupervised methods for hallucination detection.

Transportation Data Analyst

Sept. 2024 – Aug. 2025

City of Toronto

- Statistical consulting, building spatiotemporal models, and developing imputation strategies for data on networks.

OTHER SELECTED RESEARCH EXPERIENCE

Research Assistant (SMC and Posterior Inference on LLMs)

May 2024 – Dec. 2024

Advisor: Roger Grosse (University of Toronto, Vector Institute)

Research Assistant (Mathematical Biology)

May 2023 – Dec. 2023

Advisors: Juxin Liu (University of Saskatchewan), Lin Wang (University of New Brunswick)

Including only paid research experience which has not been stated elsewhere in this academic record.

TEACHING EXPERIENCE

Teaching Assistant

Sept. 2023 – May 2026

University of Toronto

- TA for MAT246H1: Concepts in Abstract Mathematics (Fall 2023, Winter 2024, Fall 2025, Winter 2026)
- TA for MAT223H1: Linear Algebra I (Fall 2024, Winter 2025)
- TA for STA130H1: An Introduction to Statistical Reasoning and Data Science (Winter 2026)
- Combination of preparing and teaching tutorials, grading assignments and tests, and holding office hours.

AWARDS & ACHIEVEMENTS

- CSSC 2nd Place Oral Research Presentation – Bachelor's Division (2026)**
Awarded for the second-best research oral presentation at the Canadian Statistics Student Conference (CSSC).
- Dean's List Scholar (2022, 2023, 2024, 2026)**
Awarded to students under the Faculty of Arts and Science at the University of Toronto who have a cumulative GPA of at least 3.50, after completing five credits.
- Computer Science Engagement Award (2025)**
Awarded by the Department of Computer Science. Recognises students who connect and apply their CS knowledge or skills in creative and innovative ways. Awarded for my paper presented at SIGSCE Virtual.
- NSERC Undergraduate Student Research Award (2024)**
Awarded by the University of Saskatchewan for a project on mathematical modelling of disease spread with differential equation models.
- University of Toronto Excellence Award (2024) – Declined**
Awarded by the Department of Statistical Science for supporting supervised research under Professor Nancy Reid.
- Samuel Beatty Scholarship (2023)**
Awarded to students studying Computer Science, Mathematics, Statistics, or Physics at the University of Toronto based on academic performance during the previous year or financial need.
- Chancellor's Scholarship (2022)**
Awarded for high academic achievement in the 2021-2022 academic year by Trinity College, University of Toronto.
- University of Toronto Scholars Award (2021)**
Awarded to outstanding secondary school applicants to the University of Toronto, on the basis of their mark average at the time of admission.
- Competitive Debate Achievements**
2-time provincial representative at national tournaments; 7-time medallist in civic/provincial tournaments.

GRANTS

- CSSC Travel Award (2025)**
Awarded to select participants at CSSC 2025 to travel to Saskatoon, SK to present their work.
- Trinity College Experiential Fund (2025)**
Awarded to students at Trinity College, University of Toronto to support experiential learning opportunities.
- Department of Computer Science Academic Travel Grant (2025)**
Awarded to students under the Department of Computer Science to attend a conference, prioritising students whose

research projects contribute to equity, diversity, inclusion, and belonging in computer science.

- **Arts and Sciences Student Union Travel Grant (2025)**

Awarded to full-time students at the University of Toronto who are presenting their work at a conference.

ACADEMIC SERVICE

- Associate Editor: *Arbor: The U of T Interdisciplinary Undergraduate Research Journal* (2025-2026, Volume 7)
- Peer Reviewer: *Machine Learning for Health Symposium (ML4H) 2025*, *Language Resources and Evaluation (LREC) 2026*
- Abstract Reviewer: *2025 Innovations in Education Conference*

ACADEMIC COMMUNITY INVOLVEMENT

Conference Volunteer

May 2025

SSC Annual Meeting 2025

- Logistical support for running the conference and supporting conference presenters with their needs.

Birds of a Feather Event on Multilinguality and Language Diversity

Jun. 2024

North American Chapter of the Association for Computational Linguistics (NAACL) 24

- Co-hosted an event at a top CS conference, leading discussion on representation of low-resource languages in NLP.
- Involved 40+ conference attendees, including leading researchers in NLP and computational linguistics.

MEDIA COVERAGE

U of T undergraduate students explore the use of AI to treat speech disfluency

Feb. 2024

By Selah Katona – UofT News

TECHNICAL SKILLS

- Languages: Python (Torch, Pandas, Scikit-Learn, psycopg), R (tidyverse, ggplot2), Java, C, SQL, JavaScript.
- Other: QGIS, Jupyter, Tableau, GitHub, Anaconda, SLURM, Microsoft Excel, Wolfram Mathematica, Latex.